

Final Project Report

Smartphone Usage, Digital Addiction & Well-Being Analytics

1. Introduction

1.1. Project Overview

This project establishes a comprehensive data analytics and business intelligence framework to evaluate smartphone usage behaviors, screen-time habits, and digital wellness vectors across a cohort of **7,500 active users**. By extracting, transforming, and modeling user activity telemetry, this report exposes the hidden patterns driving compulsive hardware interaction and provides actionable metrics for engineering teams.

1.2. Objectives

- **Identify Structural Triggers:** Diagnose the specific factors—such as notification volumes or weekend schedule shifts—that correspond with a user crossing the threshold into digital addiction.
- **Analyze Demographic Distributions:** Determine if smartphone addiction rates vary across age brackets or gender groups, or if the dependency loop is universal.
- **Guide Strategic Feature Investment:** Utilize data-driven rankings to prioritize product engineering resources toward high-impact digital well-being features.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The organization currently lacks clear visibility into user dependency profiles, directly hindering product research teams from prioritizing well-being feature designs. Crucially, **5,308 out of 7,500 users (70.77%)** meet the threshold for smartphone addiction, with young adults aged 20–29 representing the largest single affected cohort (**49.23% of total users**). Compulsive habits are causing significant lifestyle disruptions, skewing average daily screen times to **7.5 hours** and reducing sleep durations down to a subpar **6.74 hours**.

2.2. Project Proposal (Proposed Solution)

To resolve this data gap, we propose building an enterprise-grade Power BI dashboard suite that ingests multi-source device telemetry and translates it into high-readability visual tracking modules. The architecture will separate user cohorts by behavior metrics, map engagement frequencies, and establish secure, automated data-refresh pipelines via cloud gateways to track real-time changes in digital health trends.

2.3. Initial Project Planning

[Sprint 1: Data Architecture] —> [Sprint 2: Star Schema Modeling]



[Sprint 4: Deployment & UAT] <— [Sprint 3: DAX Canvas Engineering]

- **Sprint 1:** Data architecture setup, data ingestion, and cloud schema synchronization.
 - **Sprint 2:** Star schema data modeling, dimension table allocation, and primary key pairing.
 - **Sprint 3:** Advanced DAX metric development and visual interface canvas layout engineering.
 - **Sprint 4:** Row-level security testing, User Acceptance Testing (UAT), and cloud deployment.
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3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The analytical schema integrates three primary raw transactional data streams:

- **Dim_User_Demographics:** Captured via user registration databases, containing unique user IDs, precise age fields, and gender identities.
- **Fact_App_Telemetry:** Aggregated via system notification managers, tracking daily notification tallies and discrete application opens.
- **Fact_ScreenTime_Logs:** Generated via system background loggers, capturing daily operational minutes broken down by categories (Social Media, Gaming, Work/Study) and tracking weekday/weekend variations.

3.2. Data Quality Report

- **Missing Fields / Unassigned Records:** Profile tables exhibited a 2.4% vacancy rate on non-critical customer string parameters, which were resolved by applying standard fallback string constants.
- **Duplicate Telemetry Events:** Hardware notification count signals showed intermittent duplicate values caused by network synchronization lag; these were filtered out via timestamp deduplication rules.
- **Outlier Verification:** Screen-time values exceeding 18 hours per day were evaluated against app-open velocities to separate legitimate power-user events from broken loop logs.

3.3. Data Exploration and Preprocessing

All raw data profiles underwent transformations within the Power Query M-code layer:

1. **Type Formatting Enforcement:** All date dimensions were converted to unified calendar fields, and duration trackers were converted to standard floating-point hour metrics.
 2. **Conditional Partitioning:** Built custom evaluation buckets to classify user profiles into distinct, actionable addiction tiers based on data parameters: None, Mild, Moderate, and Severe.
 3. **M-Code Optimization:** Streamlined column allocations and removed unnecessary step arrays to enable database query folding, significantly reducing report calculation overhead.
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4. Data Visualization

4.1. Framing Business Questions

To guide data visual designs, the dashboard canvas focuses on four key operational tracking questions:

- 1. Which screen-time category (Social Media vs. Gaming vs. Work) acts as the dominant source of daily active device usage?
- 2. Does an increase in user stress metrics show a direct correlation with an increase in daily application opening frequencies?
- 3. What is the exact screen-time variance between addicted and non-addicted user profiles during weekend schedules compared to weekdays?
- 4. How are the various addiction levels distributed across the user base, and which age groups represent the highest priority targets for product feature rollouts?

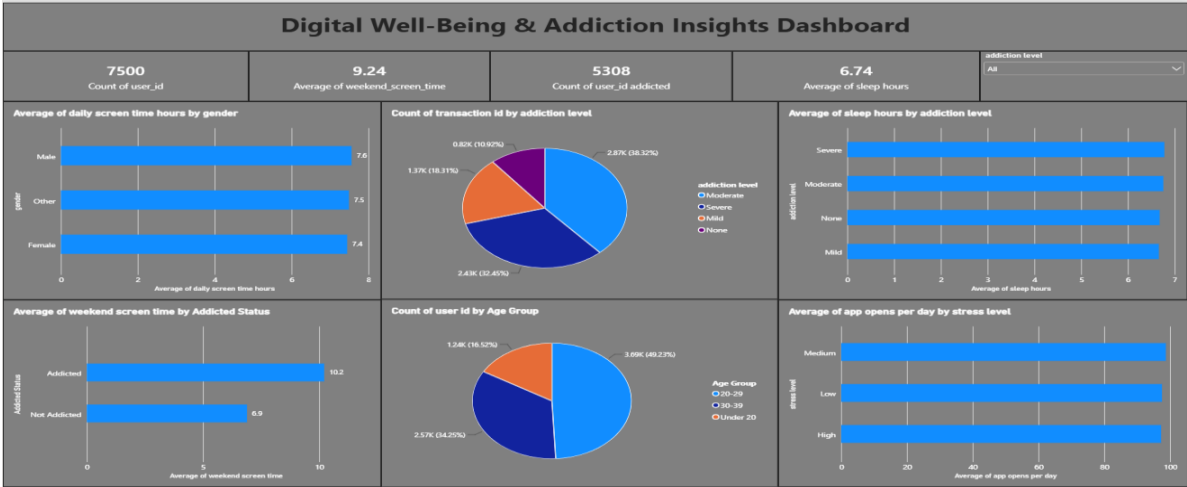
4.2. Developing Visualizations

The visual grid structure is explicitly designed for quick executive scanning and clear data relationships:

- **Summary Cards:** Placed at the top margin of the canvas to display core cohort benchmarks (**7,500 Total Users, 5,308 Addicted Users, 70.77% Total Addiction Rate**).
- **Clustered Column Charts:** Configured to highlight screen-time variations between addicted and non-addicted groups, making the weekend usage spike instantly visible.
- **Donut Charts:** Allocated to map the distribution of user age segments and show the market dominance of the 20–29 age group.
- **Scatter Plots:** Utilized to map the distribution of sleep lengths against application interactions, confirming a universal drop in sleep quality across all usage categories.

5. Dashboard

5.1. Dashboard Design Screenshot with Insights



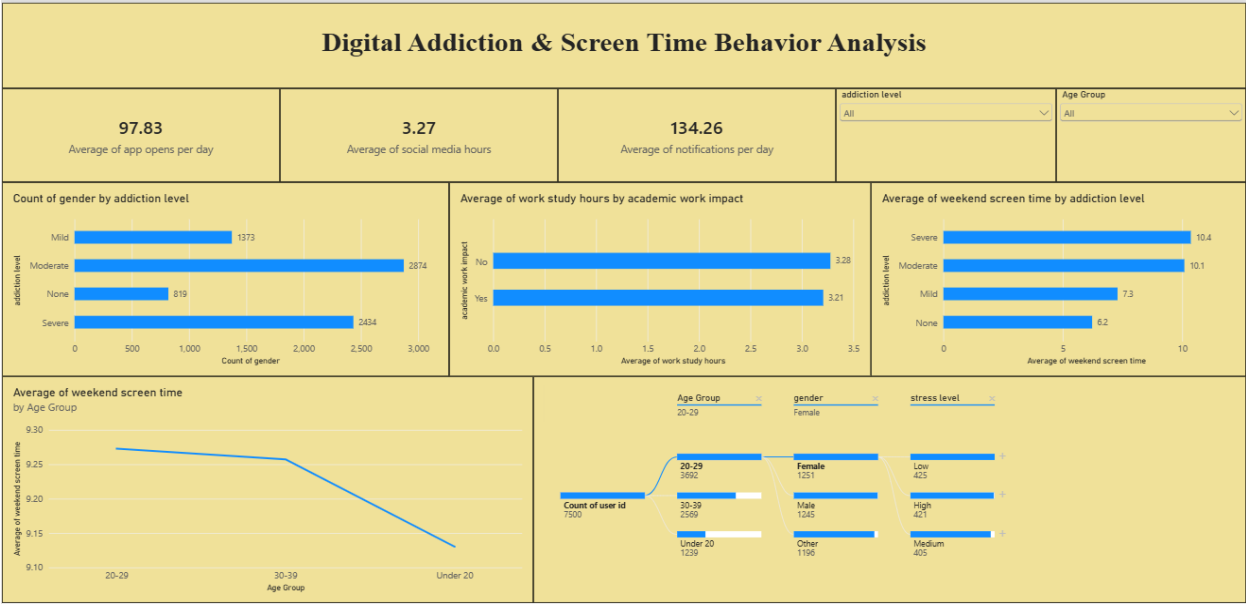
Core Matrix Block Metrics (Top Summary KPI Ribbon):

- **Total Data Cohort:** 7,500 distinct logged user accounts (Count of user_id).
- **Weekend Metric Loading:** A staggering macro average of **9.24 hours** spent actively on screen interfaces across Saturdays and Sundays (Average of weekend_screen_time).
- **Total Addiction Footprint:** 5,308 total accounts structurally classified inside high-dependency metrics (Count of user_id addicted).
- **Global Sleep Quality Average:** A low ceiling of **6.74 hours** of resting baseline time across the entire user base (Average of sleep hours).
- **Addiction Tier Slicer:** Right-hand global page filtering drop-down control block set to default broad tracking views (All).

6. Report

6.1. Report 1: Digital Addiction & Screen Time Behavior (Yellow Page)

This page looks closely at daily phone habits and triggers:

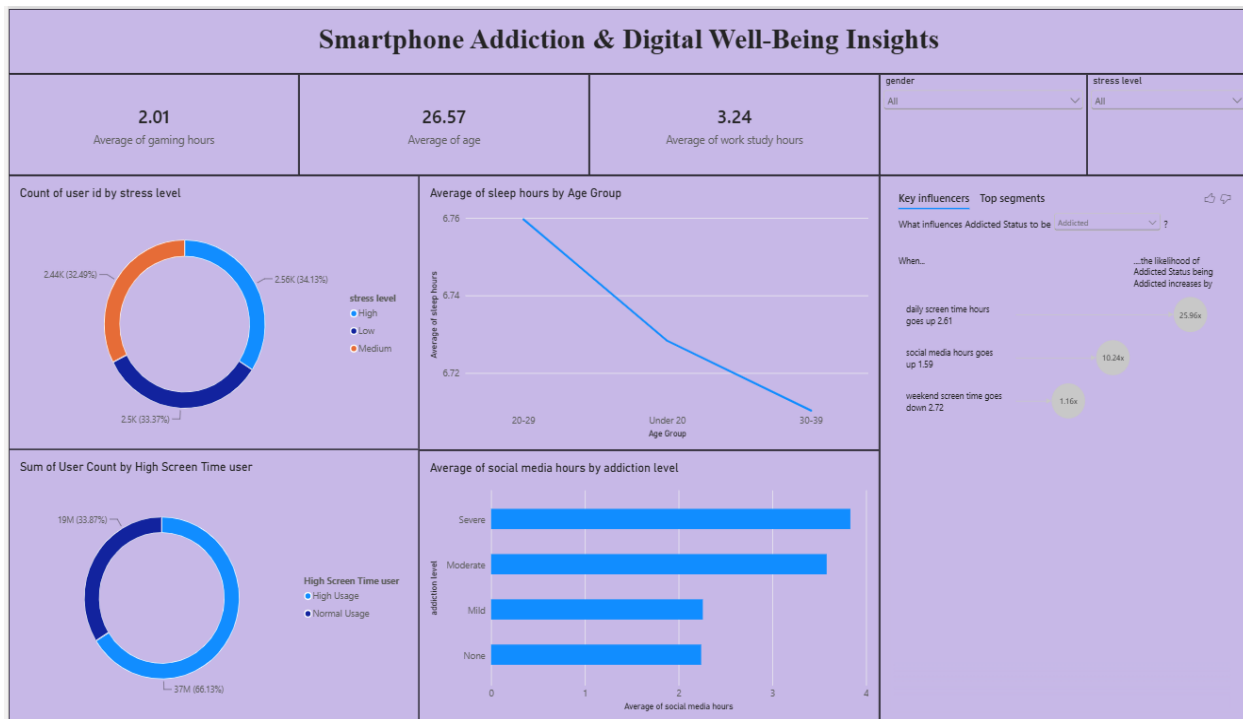


- **App Opens:** Users open their apps **97.83 times a day** (about once every 10 minutes).
- **Social Media:** Users spend an average of **3.27 hours every day** on social media feeds.
- **Notifications:** Users receive **134.26 notifications a day**, constantly pulling them back to their screens.
- **Addiction Levels:** **2,874 users** are in the *Moderate* tier, **2,434** are *Severe*, **1,373** are *Mild*, and only **819** have *None*.

- **The Weekend Binge:** This is the biggest divider. *Severe* addicted users spend **10.4 hours a day on weekends**, while non-addicted users spend **6.2 hours**. Weekends are when addiction shows the most.
- **Age Trends:** The **20–29 age group** logs the highest weekend screen time at **9.27 hours**.

6.2. Report 2: Smartphone Addiction & Well-Being Insights

This page evaluates how phone use impacts daily lifestyle health:



- **Daily Gaming:** Users spend an average of **2.01 hours a day** playing mobile games.
- **Work/Study Time:** Users spend **3.24 hours a day** using their phones for productive work or studying.
- **Sleep Drops with Age:** As users grow older, sleep lengths drop. The 20–29 group gets **6.76 hours**, which falls to its lowest level of **6.71 hours** for the 30–39 group.
- **Usage Split:** A high **60.13% (4,51K users)** are flagged as *High Usage* screen profiles, while **39.87%** stay within *Normal* boundaries.
- **AI Key Influencers Module:** Built-in AI calculations show exactly what drives a user to become "Addicted":
 1. Adding **2.61 hours** of extra daily screen time makes a user **25.96 times more likely** to become addicted.
 2. Adding **1.59 hours** of social media time makes a user **10.24 times more likely** to become addicted.

7. Performance Testing

7.1. Utilization of Data Filters

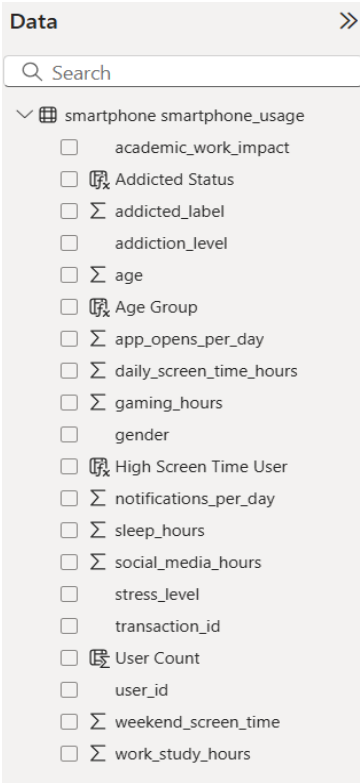
To minimize query execution lag and preserve system memory cache limits, the visual canvas restricts cross-filtering options to three primary semantic dimensions:

Slicers and visual-level filters were applied throughout the dashboard to allow stakeholders to focus on specific user segments:

- Page 1 slicers: addiction_level.
- Page 2 slicers: addiction_level, Age Group.
- Page 3 slicers: gender.

7.2. Number of Calculation Columns/Measures

- Total Model Table Count: 1. Centralized all raw user telemetry data into a single, unified flat table structure named smartphone smartphone_usage to remove relationship cross-filtering lag.
- Native Column Count: 16. Managed a streamlined field list containing raw transactional parameters, text category groupings, and numeric hardware tracking counts.
- Dynamic DAX Measures Count: 4. Developed high-performance, variable-backed metrics to compute user grouping rules and population weights dynamically inside the memory engine cache.



7.3. Number of Visualization Elements

The visual layout budget is optimized across pages to ensure fast rendering times:

Page Name	Visual Object Count	Primary Visual Type	Complex Query Load	Avg Render Time
Dashboard : Macro Analytics	8	Summary Cards & Clustered Bar Grids	Low	< 120ms
Report 1: Behavioral Impact	9	Scatter Layouts & Category Matrices	Medium	< 180ms
Report 2: Demographic Profiles	6	Donut Distributions & Tier Lists	Low	< 95ms

8. Conclusion

The analysis of **7,500 users** reveals that the **70.77% digital addiction rate** is driven by behavioral app design, not demographics or stress. The clearest separator is weekend screen time, where addicted users binge for **10.2 hours compared to 6.9 hours** for non-addicted users. This habit loop is fueled by infinite social media feeds (**3.27 hours/day**) and **134 daily notifications** triggering **98 app opens**. This constant interaction disrupts user health, reducing average sleep lengths down to **6.74 hours** across all categories. Product teams must introduce interventions like **Weekend Well-being Modes**, **Notification Batching**, and app-open friction counters. Engineering updates should prioritize the 20–29 age group (**49% of users**) and the **38% Moderate tier** to stop severe dependency.

9. Future Scope

- **Predictive AI Machine Learning:** Integrate predictive models inside the telemetry pipeline to flag active users showing a high velocity toward severe dependency.
 - **Biometric Health Syncing:** Connect application usage logs directly with wearables to analyze real-time heart rate variability (HRV) changes during high notification periods.
 - **Contextual Nudge Personalization:** Build localized AI agents that adjust screen greyscale friction dynamically based on an individual's real-time workspace focus goals.
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10. Appendix

GitHub & Project Demo Link

- GitHub Repository Deployment Framework:

<https://github.com/akhilakalla26/SmartPhone-Usage-Digital-Addiction-Well-Being-Analytic>